

9/15 ch.1.3

## \* Computer component

**Hardware**: Today have four hardwares under von neumann models  
I/O devices, processing devices and memory (storage)

**Data**: Von neumann model defined a computer as processing data and I/O data  
**Storing data**: Electrical signal is the best way to store data for electronic computing devices. We have to change our numbering system from

ch.1.2

1. Input the first number into memory.
2. Input the second number into memory.
3. Add the two together and store the result in memory.
4. Output the result.

Program

Ch.2.1

**Posicional Number System (位值進位制)**: value represent=  
$$n = \pm S_{k-1} \times b^{k-1} + S_{k-2} \times b^{k-2} + \dots + S_1 \times b^1 + S_0 \times b^0$$
  
 $S$  is the symbol (number),  $b$  is the base (radix)

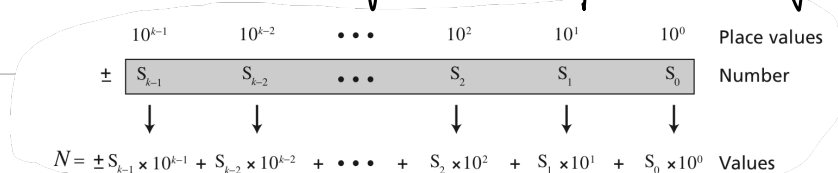
**Decimal system (十值進位系統)**: the base is 10 and the symbol is the set zero to nine

\* In Decimal system, the number is written as  $(522.30)_{10}$ . but we dropped the parentheses, the base, the plus sign, thus we simplify it as  $522.30$

**Integers**: An integral numbers without fractions, written as:

$$N = \pm S_{k-1} \times 10^{k-1} + \dots + S_0 \times 10^0$$

**Place Value**: A number system to represent integer



|      |               |               |               |               |         |
|------|---------------|---------------|---------------|---------------|---------|
| e.g. | 7             | 2             | 4             | 0             |         |
|      | $\times 10^3$ | $\times 10^2$ | $\times 10^1$ | $\times 10^0$ |         |
|      | 17000         | + 200         | + 40          | + 0           | = 17240 |

Maximum Value / Minimum Value